



AirResolve AI

Customer-Facing Airline Resolution Agent

An agentic AI prototype for automated airline disruption resolution —
grounded, deterministic, explainable

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Assignment

[LIVE DEMO: STREAMLIT.APP](#)

[GITHUB: MANUJSHARMA03](#)

The Problem

The Repetitive Loop

- Airline disruptions flood support queues — especially during weather or operational events
- Human agents repeat the same 3 steps on every ticket: **look up booking → check policy → explain outcome**
- Repetitive at scale = slow, costly, and error-prone

Two Critical Failure Modes

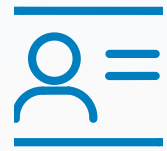
- **Under-delivering:** Agent misses legitimate entitlements — e.g., forgets lounge pass at a 4h delay
- **Over-promising:** Agent invents unauthorized compensation — e.g., free upgrade — costing the airline and creating inconsistency



Need: a fast, consistent system that **never exceeds its authority**

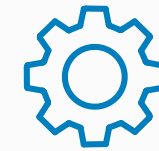


Our Solution: AirResolve AI



Identify & Retrieve

Recognize customer via PNR, name, or flight number — pull real booking data, never fabricated



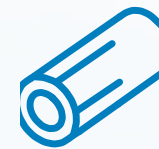
Deterministic Policy Engine

Rules-based decisions — agent states exactly which policy applies and why, every single time



Act or Escalate

Execute eligible actions automatically; hand off judgment calls to humans immediately

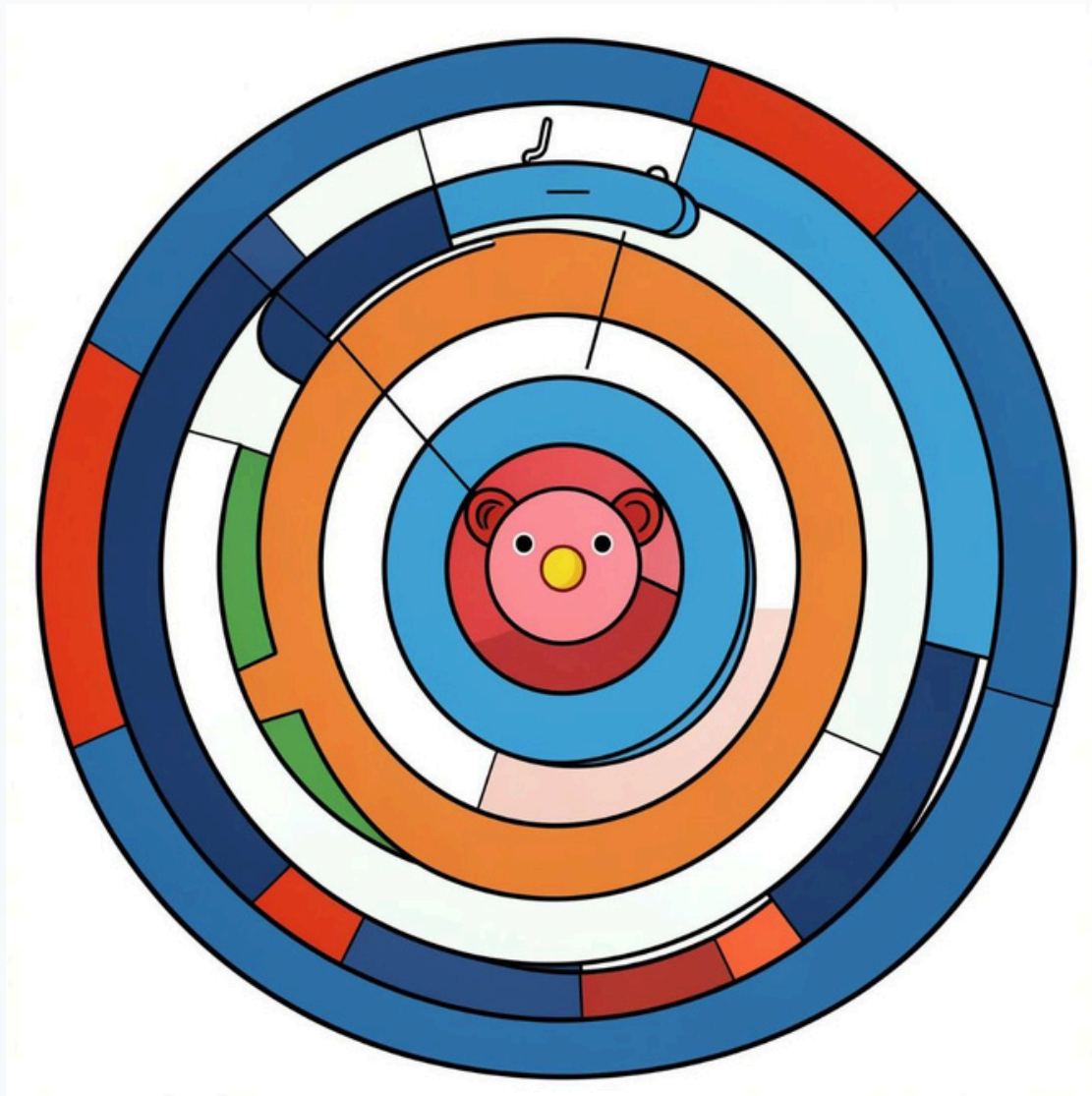


Full Audit Trail

Every identification, retrieval, policy match, and action logged with timestamp for compliance

📄 Design philosophy: **Automate the repeatable, rules-based 80%** — keep humans in the loop for judgment calls

Objectives



1 Resolve Correctly & Consistently

Regardless of customer tone or phrasing

2 Zero Hallucination Tolerance

Never invent policy, compensation, or facts — on any decision

3 Always Defer Out-of-Authority

Upgrades, large waivers, legal threats, formal complaints → human agent

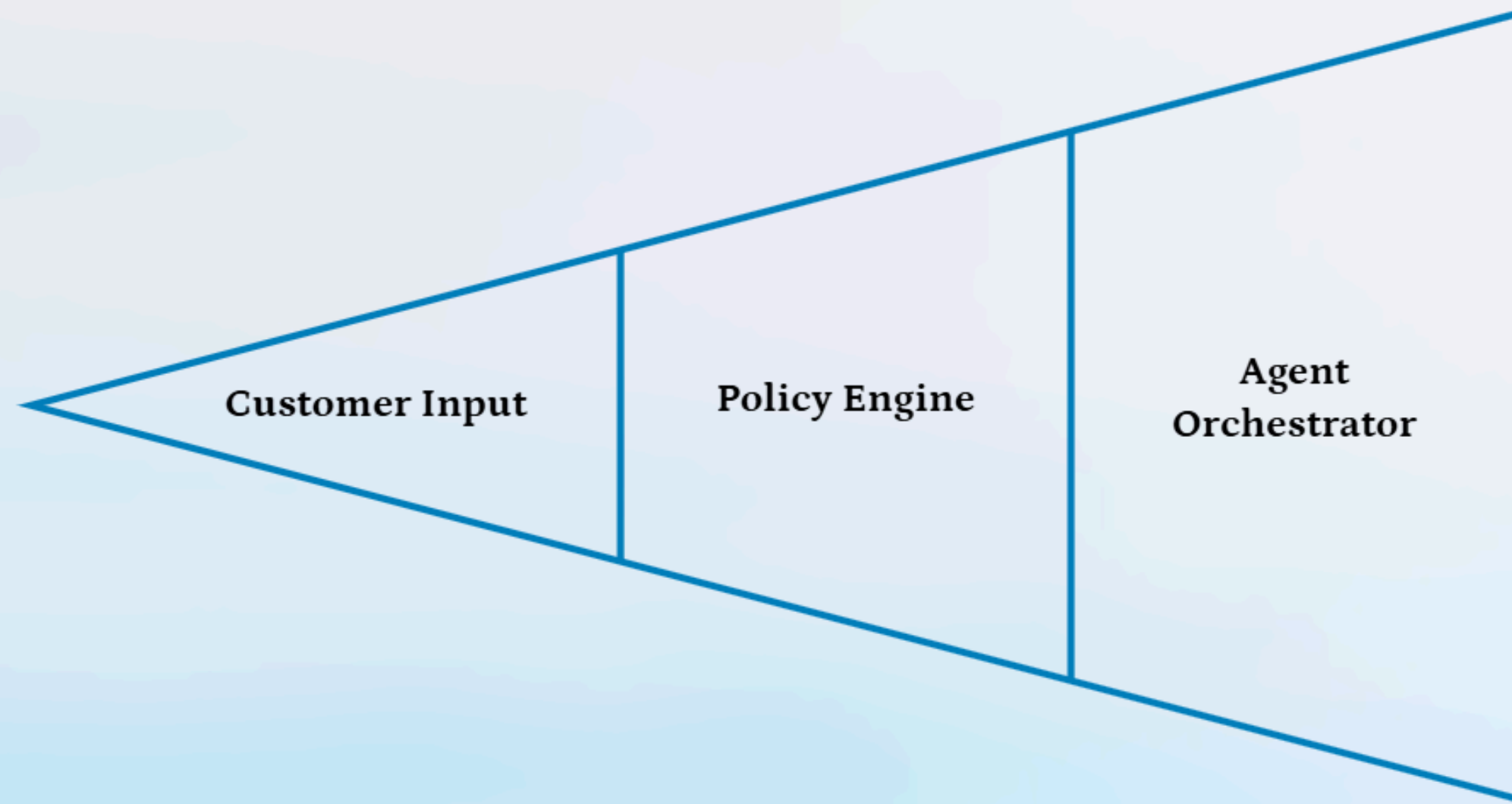
4 Explainable & Auditable

No black-box outcomes — every decision traceable

5 Demo-Ready in 15 Minutes

Pre-loaded scenarios for instant live walkthrough

System Architecture



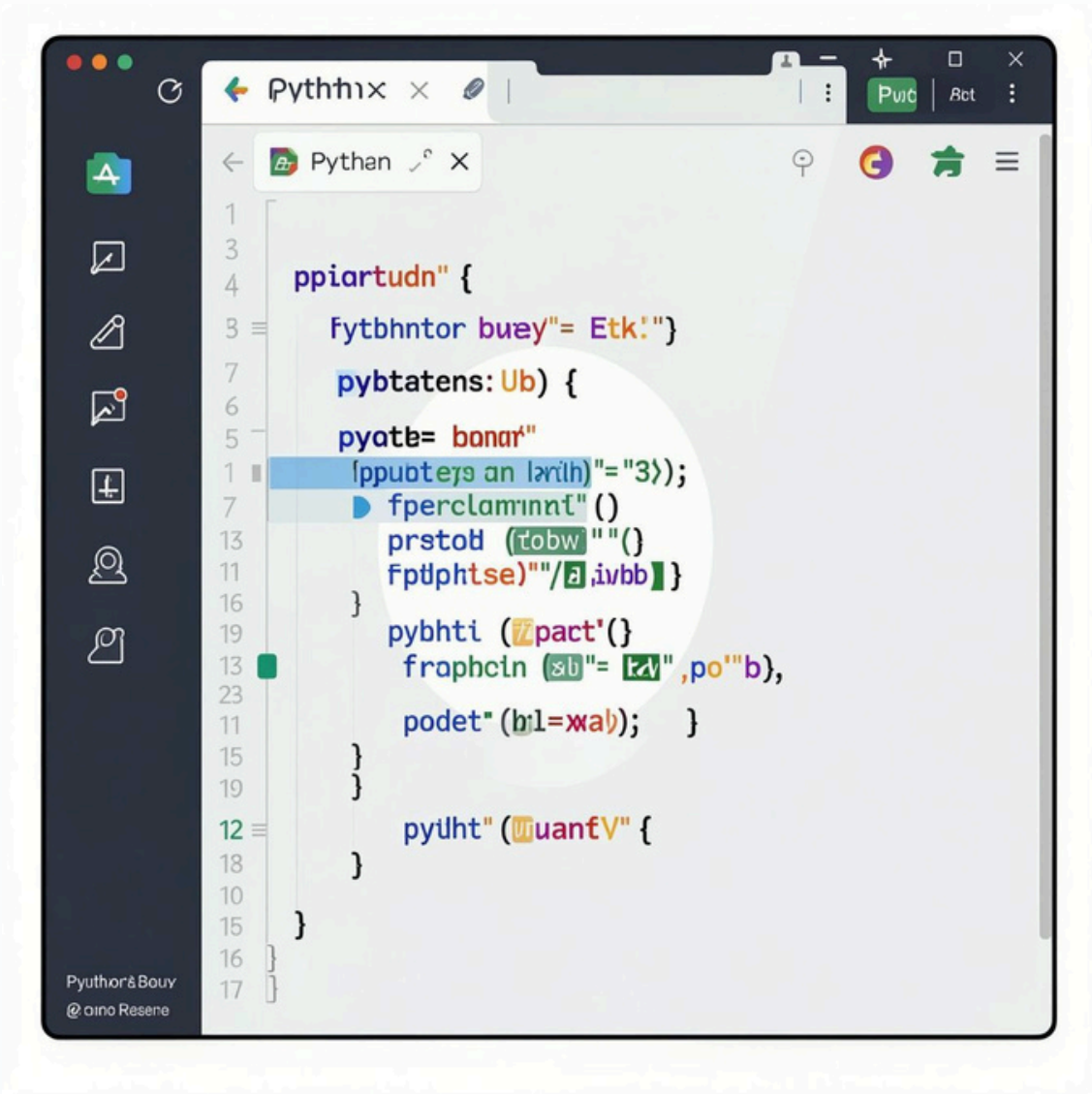
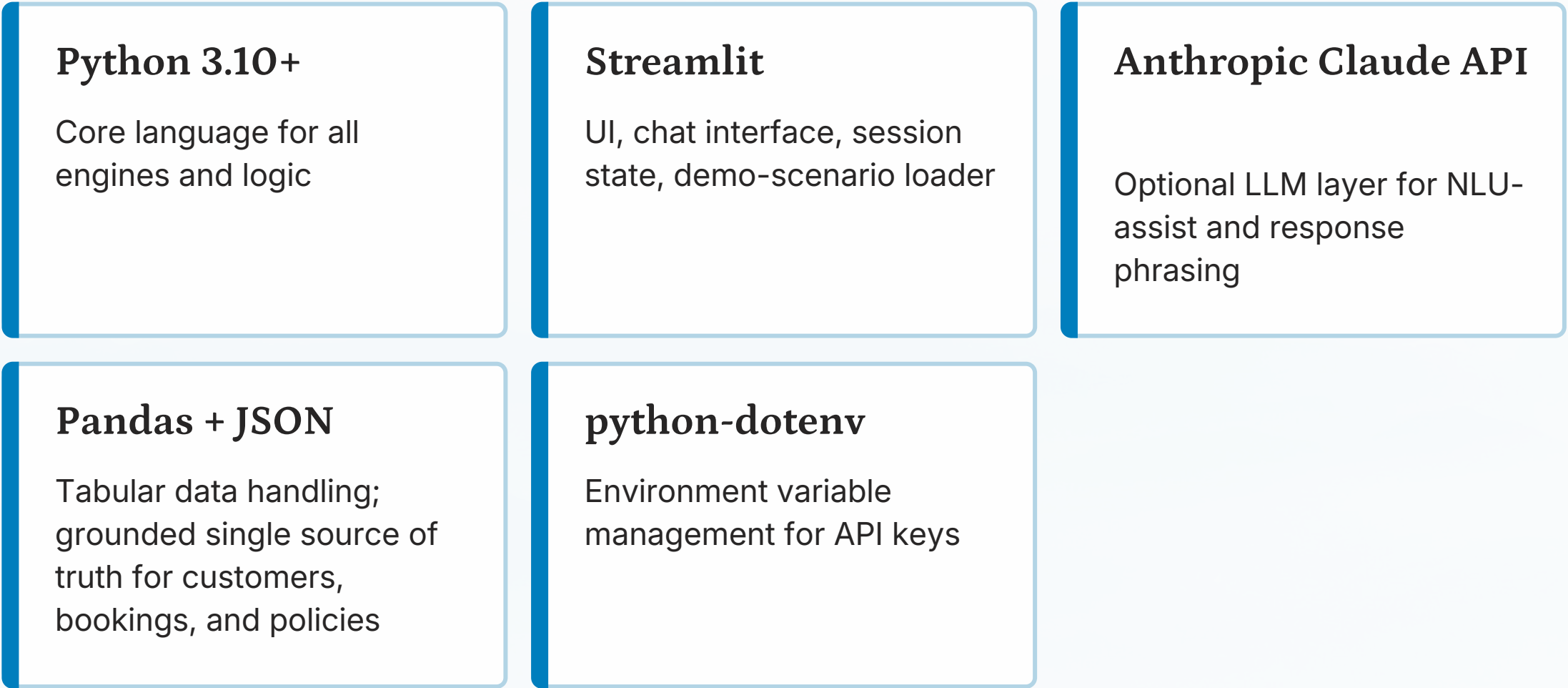
LLM Boundary — Critical Design Decision

- LLM used **only** for intent/sentiment detection and natural-language phrasing
- LLM **never** decides eligibility, compensation amounts, or escalation triggers
- Policy engine functions take **zero LLM input** — no code path for model to influence outcomes

"Safe-by-Construction" Design

- Structurally enforced separation between language model and decision logic
- Audit-safe architecture for regulated industries
- Deterministic policy outcomes regardless of LLM availability

Technology Stack



✔ **Mock-Mode Design:** App runs fully with zero API key configured — policy decisions behave identically; only response phrasing falls back to deterministic templates. Safe to demo publicly.

How the Agent Works: 7-Step Pipeline



Intent Detection

Keyword detector finds CANCELLATION, REFUND, HOTEL, UPGRADE, LEGAL_THREAT + sentiment; LLM pass adds intents, never removes rule-based ones



Identification

Recognizes customer via PNR, name, or flight number; reuses memory; one clarifying question if needed



Retrieval

Pulls real booking and flight record from data/*.json — never fabricated



Policy Engine Decision

Deterministic functions decide eligibility, allowed actions, reason, and escalation requirement



Action or Escalation

Eligible actions executed via action_engine.py; out-of-authority requests escalated via escalation.py



Response Generation

LLM or deterministic template turns decision into a natural, empathetic reply grounded in JSON data



Audit Trail

Every step logged with timestamp — identification, retrieval, policy match, action, and escalation

Policy Engine & Escalation Rules

✓ Allowed Actions

Scenario	Entitlement
Cancellation	Free rebooking within 24h OR full refund
Delay ≤3h	Meal voucher
Delay >3h	Meal voucher + lounge access
Delay >5h	Meal + lounge + hotel (delayed hours only)
Refund	Full refund within 7 business days to original payment
Fare Difference	Agent waives up to ₹1,500; customer pays remainder
Gold/Platinum Tier	Priority rebooking only — no additional compensation for tier

✗ Escalation Triggers

Compensation beyond policy limits	Fare-difference waiver above ₹1,500
Exception for non-airline-caused disruption	Legal threats or formal complaint intent
Refund to a different payment method than original	Any request policy doesn't explicitly grant (e.g., free upgrade)

Demo Scenarios & Results

Customer	Scenario	✓ Granted	× Escalated
Priya Nair Gold · PNR SK4821X	Flight SK-204 cancelled; demands full refund + free business upgrade	Refund processed automatically	Upgrade escalated — no policy grants free upgrades
Arvind Kulkarni Silver · PNR TR1190B	Flight SK-118 delayed 4h; asks for hotel	Meal voucher + lounge granted (>3h delay)	Hotel denied — below 5h threshold
Meher Kaur Platinum · PNR WL7742	Flight SK-305 delayed 6h; wants full-night hotel + ₹2,000 fare waiver	Meal + lounge + hotel granted (delayed hours only)	₹2,000 waiver escalated — exceeds ₹1,500 authority

 All three scenarios pre-loaded in sidebar for instant live walkthrough — **Live Demo:** streamlit.app

Limitations, Future Work & AI Tools Used

Current Limitations

- Prototype only — all actions (refunds, vouchers, hotel bookings) are **simulated**
- Only three supplied customers, bookings, and policies; anything outside is explicitly out-of-scope
- Keyword-based intent/sentiment detection with optional LLM assist — not a full NLU model
- Single-process, in-memory session state — no persistent database or multi-user backend yet

AI Tools Used

- **Claude (Anthropic)** — architecture, logic, and documentation
- **Codex/GPT (OpenAI)** — code generation and debugging
- All AI-generated code reviewed, tested, and adapted by author

Future Improvements

- **Real Backend Integrations**
Live PNR systems, payment gateways, hotel and voucher APIs
- **Proper NLU Model**
Confidence scores and human-in-the-loop review for ambiguous inputs
- **Multi-Turn Slot-Filling**
Handle partially specified requests across multiple conversation turns
- **Supervisor Dashboard**
Role-based interface to action and resolve escalation queue items
- **Persistent Storage & Multi-Language**
Real database for audit trail, case history, and broader customer reach